22ITRTO - AI for cybersecurity

Assignment - II

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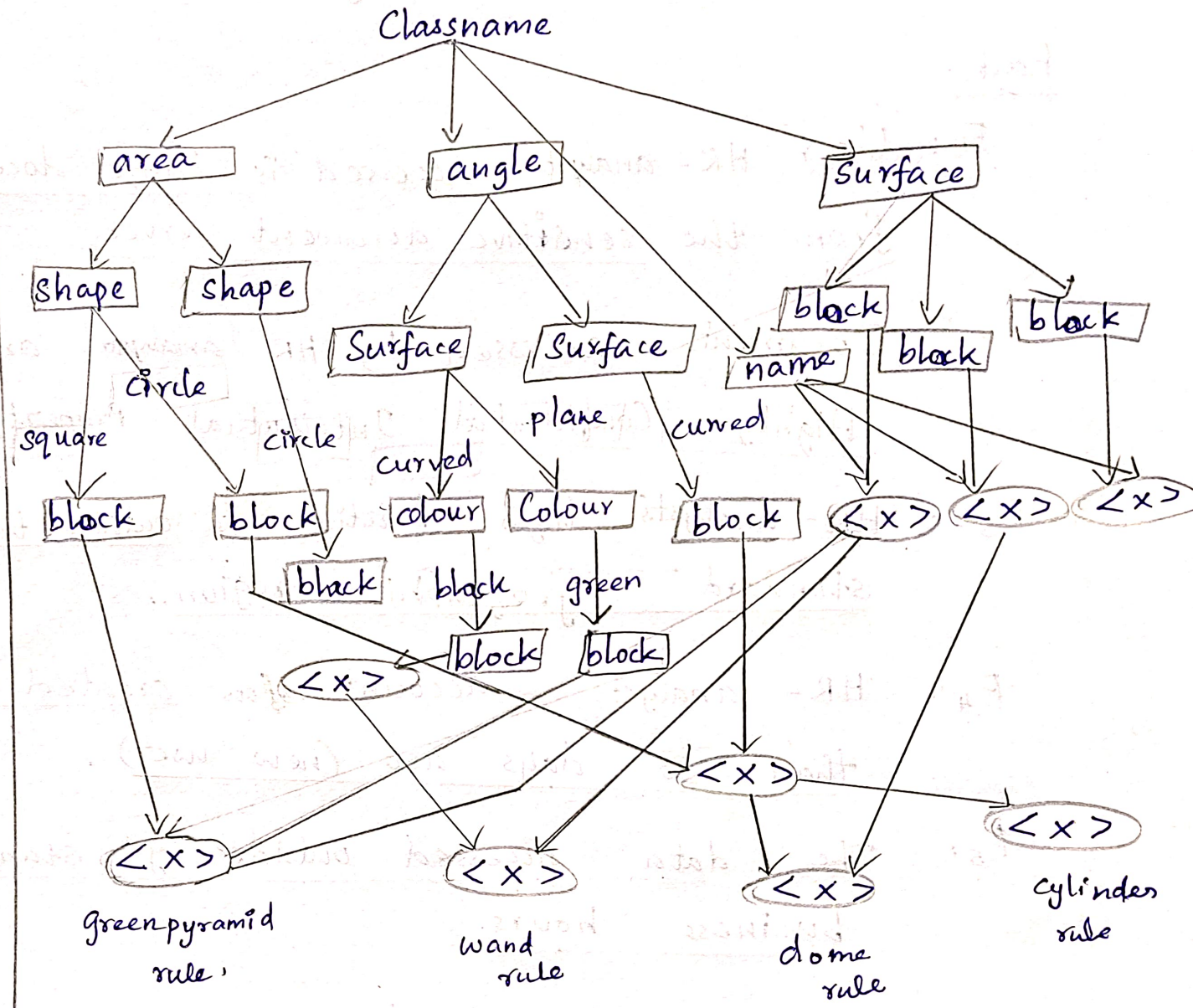
22IT036

① Given the facts and Rules form the RETE graph.



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② Given the facts and rules, apply forward Chaining and backward chaining.

Facts:

F_1 : User HR-analyst accessed is Large documents from the sensitive document server.

F_2 : Documents accessed by HR analyst are Highly Confidental Intellutual Property.

F_3 : HR-analyst's login location is outside the standard geographical region.

F_4 : HR-analysts account was created less than 30 days ago (new user).

F_5 : The data accessed outside of standard business hours.

Intermediate facts from rules:

IF_1 : Mass Data Access Detected

IF_2 : Suspicious Data Access Profile.

IF_3 : High risk Data exfiltration pattern.

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Goal : (Critical Fact)

CF : Critical Insider Threat incident

Rules into logical / symbolic form.

$R_1 : F_1 \wedge F_2 \rightarrow IF_1$

$R_2 : IF_1 \wedge F_3 \rightarrow IF_2$

$R_3 : IF_2 \wedge F_5 \rightarrow IF_3$

$R_4 : IF_3 \wedge F_4 \rightarrow CF$

Forward chaining

Facts $\Rightarrow$ F_1, F_2, F_3, F_4, F_5

Rules $\Rightarrow$ R_1, R_2, R_3, R_4

Goal $\Rightarrow$ CF & Intermediate Facts : IF_1, IF_2, IF_3

Facts	Rules Triggered	Rules Fired
F_1, F_2, F_3, F_4, F_5	R_1	R_1
F_1, F_2, F_3, F_4, F_5 IF_1	R_2	R_2
F_1, F_2, F_3, F_4, F_5, IF_1, IF_2	R_3	R_3
F_1, F_2, F_3, F_4, F_5 IF_1, IF_2, IF_3	R_4	R_4

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Backward Chaining:

Starting from the goal: CF

Facts	Goals	Matching rules
F_1, F_2, F_3, F_4, F_5	CF	R_4
F_1, F_2, F_3, F_4, F_5	IF_3	R_3
$F_1, F_2, F_3, F_4, F_5, IF_3$	IF_2	R_2
$F_1, F_2, F_3, F_4, F_5, IF_3, IF_2$	IF_1	R_1

The goal 'CF' is derived successfully from the given facts.